1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), high tumor budding at the invasive front reflects an active epithelial-to-mesenchymal transition (EMT) program as evidenced by an elevated EMT gene signature; this EMT activation mediates enhanced local invasion and early dissemination, partially driving worse overall survival independent of conventional pathologic stage and nodal status. Consequently, tumors with high TB and high EMT_signature constitute a high-risk subgroup with prognostic information beyond stage, and the TB→EMT→invasion axis represents a mechanistic link between histology and outcome in MIBC.

Element	Value
Imaging feature	Tumor budding at the invasive front
Anchor gene(s)	EMT gene signature
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

EMT gene signature: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Tumor stage (pT/pN)	Supporting modifier from knowledge graph
Secondary	Lymphovascular invasion	Supporting modifier from knowledge graph
Secondary	Tumor–stroma ratio (TSR)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective cohort analysis: assemble MIBC patients with available histopathology to score TB and with transcriptomic data to compute EMT_signature scores; perform multivariate Cox models adjusting for stage, nodal status, grade, and other relevant covariates.

Prediction 2: Interaction/mediation analysis: test whether EMT_signature mediates the TB–OS association and quantify the proportion mediated.

Prediction 3: Validation at protein level: in a subset, perform immunohistochemistry for epithelial vs mesenchymal markers (e.g., loss of epithelial adhesion and gain of mesenchymal traits) to corroborate the EMT state associated with TB.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Tumor stage (pT/pN): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.

- Lymphovascular invasion: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.